

to the requirement of every pixel.

In contrast to Microvision's single MEMS, TI's DLP uses an array of millions of MEMS sensors. For every pixel, a corresponding MEMS mirror acts as a switch to turn the pixel 'on' and 'off.' This results in extremely high quality of the image. TI's DLP technology is known for giving a high contrast, fast switching and high optical efficiency. "DLP technology, along with LEDs to illuminate the projection screen, gives almost 100 per cent true colour requirement," shares S. Rajendran, CMO of Acer.

Liquid-Crystal-on-Silicon (LCoS) technology from Light Blue Optics uses an innovative approach to projection with its miniature light engines. By using an algorithm, its holographic laser projection method converts the image into diffracted patterns or holograms that are displayed on the LCoS microdisplay. As the holograms are illuminated by coherent red, green and blue sources of laser light, a 2-D image is formed. This technique requires no moving part or projection lens, and is there-

fore suitable for miniaturisation.

## Add-ons to drive the projector

A projector in a cellphone is not just an application running on the cellphone's operating system, but projecting an image requires an optical assembly, light and, of course, the front-end software to drive the projector. LG's eXpo has the projector assembly as a separate add-on that is detachable, while Samsung phones have these components within the bar phone.

"DLP chipset, LEDs, optics and LED driver circuitry would be the major components that go into the pico module that can be embedded into a mobile phone," shares Ganesh. The LED is used for illumination.

Rajendran adds, "There has to be a power supply for the illumination part and a logic board to understand the signals and project them as an image." It's all solidstate electronics without any moving part. "Integrating the lens assembly is not an issue, as cellphones already have cameras embedded in them."

The projector GUI helps to import videos, images and presentation to be fed to the projector hardware. Rajendran shares, "Smart phone utility for projectors is a software that can run on your small phone. This is used to drive external palm-size pico-projectors through your cellphone."

## Market for pico projector phones

Embedded projectors are a perfect fit for space-constrained mobile devices when there is a need for projecting a

## Some Projector Phones Available Today

1. Samsung Show W7900 in Korea and i7410 in Europe
2. Samsung i8520 Halo (uses a DLP projector running on Android OS)
3. Samsung W9600 (uses TI's new WVGa DLP chip)
4. LG eXpo (with a pico-projector 'add-on')

large image. "Smartphones with embedded pico projectors are set to experience a 60-fold growth in shipments during the next four years," predicts market research firm iSuppli.

One of the early users of pico projector based cellphones are likely to be the business people. Pico projectors will come in handy for them to make quick presentation to a small group of people from their mobile devices, says the iSuppli report. These will also appeal to the general users by allowing large-sized display of their videos, Internet sites and applications.

Although there exists a great potential for acceptance of pico projector phones, there are some challenges that need to be addressed. "The growth potential for embedded pico projectors will be limited during the next few years due to challenges in areas including power consumption, size and manufacturing," observes Sanju Khatri, iSuppli principal analyst for signage/projection. As these issues are resolved, pico projectors will appear in many more mobile electronic devices.

"The pico-projector embedded in a cellphone will consume a lot of power for illumination. Power management and therefore battery drain management is a cusp that it has to overcome for it to become mainstream," says Rajendran.

Rajendran feels that projector-based phones definitely have a market as the workstyle changes to frequent interactions with small and dedicated groups. Ganesh believes that the projector will become as popular as a camera in a phone today. ●



Samsung i8520 Halo projector phone